adapted from Appendix C.2, not numbered problems

In [1]:
```
import sqlite3

con = sqlite3.connect(":memory:")
print("Connected to in-memory SQLite database")
```

```
Connected to in-memory SQLite database
```

In [2]:
```
con.execute("""
CREATE TABLE customers (
    id      INTEGER PRIMARY KEY,
    name    TEXT NOT NULL,
    city    TEXT NOT NULL,
    balance REAL NOT NULL
)
""")
print("Table created")
```

```
Table created
```

dialect awareness: MySQL to SQLite

SQLite does not enforce the length and precision limits of MySQL types like VARCHAR(50) and DECIMAL(10,2), so I used TEXT and REAL instead. REAL is floating point, so a production database would use DECIMAL in MySQL for money. MySQL uses AUTO_INCREMENT to generate ids. In SQLite, declaring a column INTEGER PRIMARY KEY does the same job: if you leave id out of an INSERT, SQLite assigns the next number automatically.

In [3]:
```
rows = [
    (1, "Ava Thompson",  "Seattle",   1250.50),
    (2, "Liam Carter",   "Portland",   980.00),
    (3, "Sofia Ramirez", "Austin",    2310.75),
    (4, "Noah Kim",      "San Diego",  455.20),
    (5, "Emma Dubois",   "Chicago",   1789.99),
    (6, "Oliver Singh",  "Denver",     620.40),
    (7, "Mia Johansson", "Boston",    3050.00),
    (8, "Ethan Okafor",  "Atlanta",    875.65),
    (9, "Chloe Nguyen",  "Phoenix",   1432.10),
    (10, "Lucas Silva",  "Miami",      299.95),
]

for r in rows:
    con.execute(
        "INSERT INTO customers (id, name, city, balance) VALUES (?, ?, ?, ?)",
        r
    )
con.commit()
print(f"Inserted {len(rows)} rows")
```

```
Inserted 10 rows
```

In [4]:
```
cur = con.execute("SELECT * FROM customers")
print([d[0] for d in cur.description])
for row in cur.fetchall():
    print(row)
```

```
['id', 'name', 'city', 'balance']
(1, 'Ava Thompson', 'Seattle', 1250.5)
(2, 'Liam Carter', 'Portland', 980.0)
(3, 'Sofia Ramirez', 'Austin', 2310.75)
(4, 'Noah Kim', 'San Diego', 455.2)
(5, 'Emma Dubois', 'Chicago', 1789.99)
(6, 'Oliver Singh', 'Denver', 620.4)
(7, 'Mia Johansson', 'Boston', 3050.0)
(8, 'Ethan Okafor', 'Atlanta', 875.65)
(9, 'Chloe Nguyen', 'Phoenix', 1432.1)
(10, 'Lucas Silva', 'Miami', 299.95)
```

In [5]:
```
print("Row count:", con.execute("SELECT COUNT(*) FROM customers").fetchone()
```

```
Row count: 10
```